Do Now · retrieval

Floating point, binary and Boolean mathematics

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is normalisation's purpose?

Q2

What is first-two-bit rule for normalised two's-complement mantissa?

Q3

State De Morgan: NOT(A AND B).

Q4

Simplify (A∧B)∨(¬A∧B) with working.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — Normalising signed binary mantissas shifts bits and compensates exponent to preserve the same value.

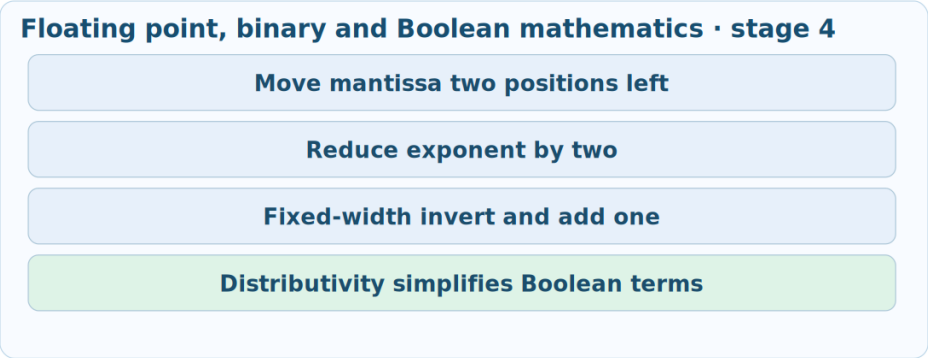
Core idea 2

Apply — For negative integer conversion, write exactly eight positive bits, invert each, then add one.

Core idea 3

Evaluate — For algebraic simplification, show each Boolean law used rather than only the final expression.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

mantissa — Significand representing precision.

exponent — Power-of-two scale.

two's complement — Representation for signed integers.

normalisation — Shift/scaling preserving represented value.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

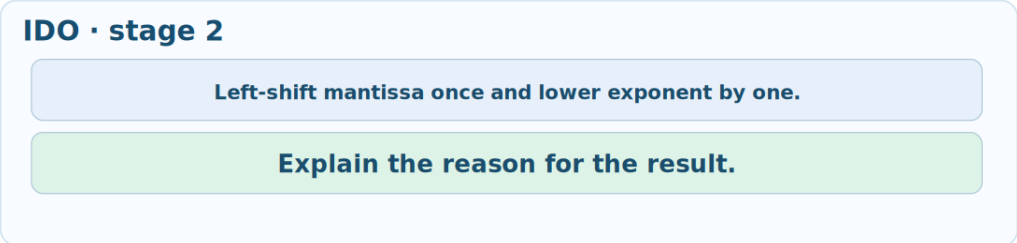
Problem

Normalise $0.0101_2 \times 2^4$.

Follow the worked method

1. Predict first — Left-shift mantissa once and lower exponent by one.
2. Explain and check — Explain the reason for the result.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

Invert all eight bits then add one, yielding 11110101.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Normalise $0.00101_2 \times 2^5$ and express its decimal value. [6]

Question 2

Convert −23 to 8-bit two's complement, showing every step. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Simplify A∨(¬A∧B) using identities; verify for A0,B1. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Normalisation shifts mantissa but never changes exponent.

Use key vocabulary from memory

1. Explain mantissa in your own words.

2. Explain exponent in your own words.

3. Explain two's complement in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.